Uncovering Heterogeneous Treatment Effects: Analysis of the LaLonde NSW (National Supported Work) Data

Comparing 8 Methods on the Lalonde (1986) Dataset

Research Question & Motivation

Does the job training program (NSW) increase earnings, and does the effect vary across individuals?

Dataset

Lalonde (1986): National Supported Work (NSW) job training experiment. 445 obs, randomized treatment.

Outcome (Y)

Real earnings in 1978 (post-treatment)

Treatment (W)

Binary: 1 = received job training, 0 = control

Covariates (X)

Age, education, race, marital status, no-degree, earnings in 1974 & 1975

8 Methods - Overview

1 OLS with Controls

Econometric

A traditional parametric regression model that assumes a linear relationship between covariates, the treatment indicator, and the outcome.

2 Propensity Score Matching (PSM)

Matching

A semi-parametric method designed to mimic a randomized experiment by pairing treated units with control units that have a similar probability of receiving treatment.

3 Inverse Probability Weighting (IPW)

Semiparametric

Instead of matching individuals, IPW reweights the entire sample to create a synthetic population where the treatment assignment is independent of the covariates

4 Augmented Inverse Probability Weighting (AIPW)

Doubly Robust

A doubly robust estimator that combines an outcome model (what Y should be) and a propensity score model (how likely W is).

5 Double Machine Learning (DML)

Semiparametric ML

To estimate treatment effects when confounding variables are highly complex or high-dimensional. Solves regularisation/overfitting bias.

6 T-Learner

ML Meta-learner

A meta-learner strategy that breaks the causal problem down into two entirely separate standard supervised machine learning tasks.

7 Causal Forest

ML Tree-based

An extension of Random Forests explicitly redesigned to optimize for finding treatment effect heterogeneity rather than just predicting an outcome.

8 Bayesian Causal Forest (BCF)

Bayesian Trees

BCF is a targeted modification of **BART (Bayesian Additive Regression Trees).** It calculate how a specific intervention impacts different individuals based on their unique characteristics

Shared Setup & Residualization (DML Pre-processing)

1

Propensity Score W

Random forest regression of W on X — used by IPW, AIPW, BCF

2

Outcome Prediction Y

Random forest regression of Y on X — used by Causal Forest

3

Residualization

5-fold cross-fitted residuals: $\tilde{Y} = Y - \hat{Y}$, $\tilde{W} = W - \hat{W}$ (DML)

4

Honest Splitting

Causal forest splits data for estimation vs. inference

Methods 1–4: Econometric & Semiparametric

1. OLS with controls

Linear regression of Y on W and X. Parametric, assumes homogeneous treatment effect. Estimates ATE

2. Propensity Score Matching (PSM)

1:1 nearest-neighbour matching on logistic propensity score. Bias-adjusted. Estimates ATT.

3. Inverse Probability Weighting (IPW)

Hajek estimator with clipped propensity scores. Reweights treated/control to form pseudo-population.

4. Augmented Inverse Probability Weighting (AIPW)

Doubly robust: combines outcome model (μ_1, μ_0) and propensity score. Consistent if either model is correct. DR score used.

Methods 5–8: ML & Bayesian

5. Double Machine Learning (DML)

Uses a 5-fold cross-fitted. OLS on $\tilde{Y} \sim \tilde{W}$. Robust to regularization bias and overfitting in ML nuisance models.

6. T-Learner

Train one Random Forest only on the people who took the training, and a second one only on the people who didn't. To find the effect for a specific person,it ask both models for a prediction and subtract the two.

7. Causal Forest

Targets heterogeneous CATEs. Uses W & Y. Calibration test available (if the "heterogeneity" it found is actually real or if the model is just seeing patterns in random noise.)

8. Bayesian Causal Forest (BCF)

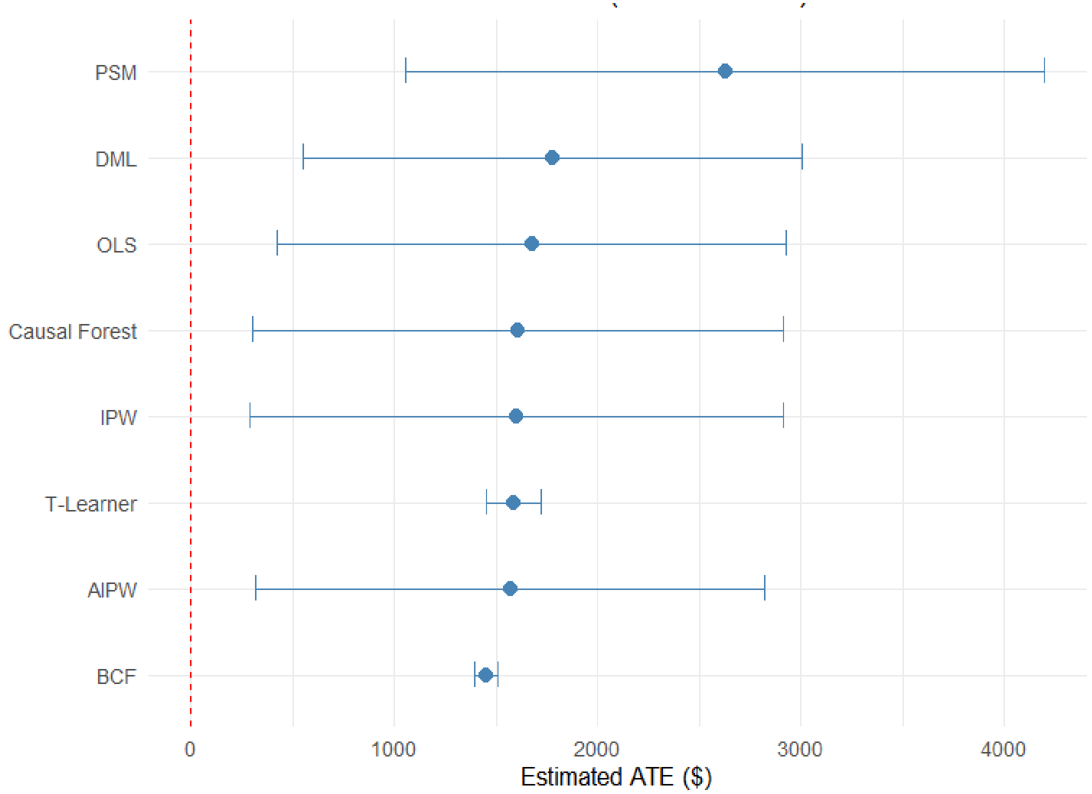
Bayesian Causal Forest (Hahn et al. 2020). Separates prognostic (μ) and treatment (τ) trees. Plugs in W to reduce confounding bias.

ATE Results — All 8 Methods

Method	ATE ($)	SE ($)	95% CI	Family
OLS	1,676	639	[425, 2,928]	Econometric
PSM	2,626	802	[1,054, 4,198]	Matching
IPW	1,599	669	[287, 2,911]	Semiparametric
AIPW	1,573	639	[321, 2,826]	Doubly Robust
DML	1,777	625	[552, 3,001]	Semiparametric ML
T-Learner	1,585	69	[1,449, 1,721]	ML Meta-learner
Causal Forest	1,591	666	[285, 2,897]	ML Tree-based
BCF	1,427	27	[1,374, 1,481]	Bayesian Trees

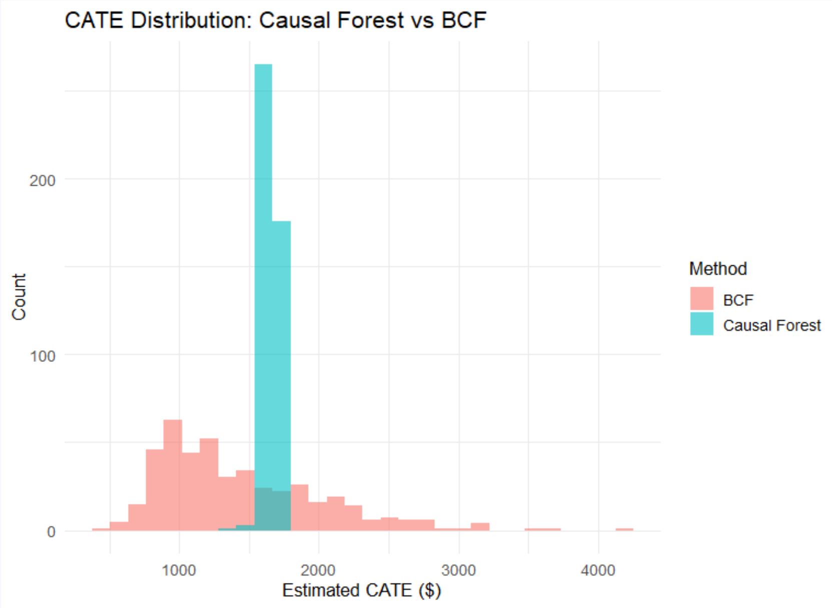
Note: PSM estimates ATT; all others estimate ATE. BCF/T-Learner SEs reflect posterior/sampling variance of CATE mean, not inference SE.

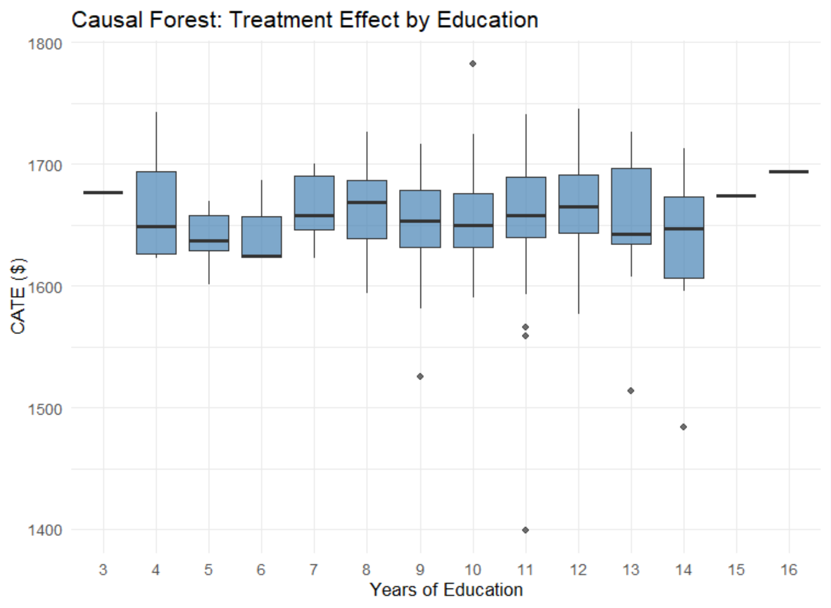
ATE Estimates — Visual Comparison



Treatment Effect Heterogeneity (CATEs)

Causal Forest & BCF both estimate individual-level CATEs. Key finding: substantial heterogeneity across individuals.





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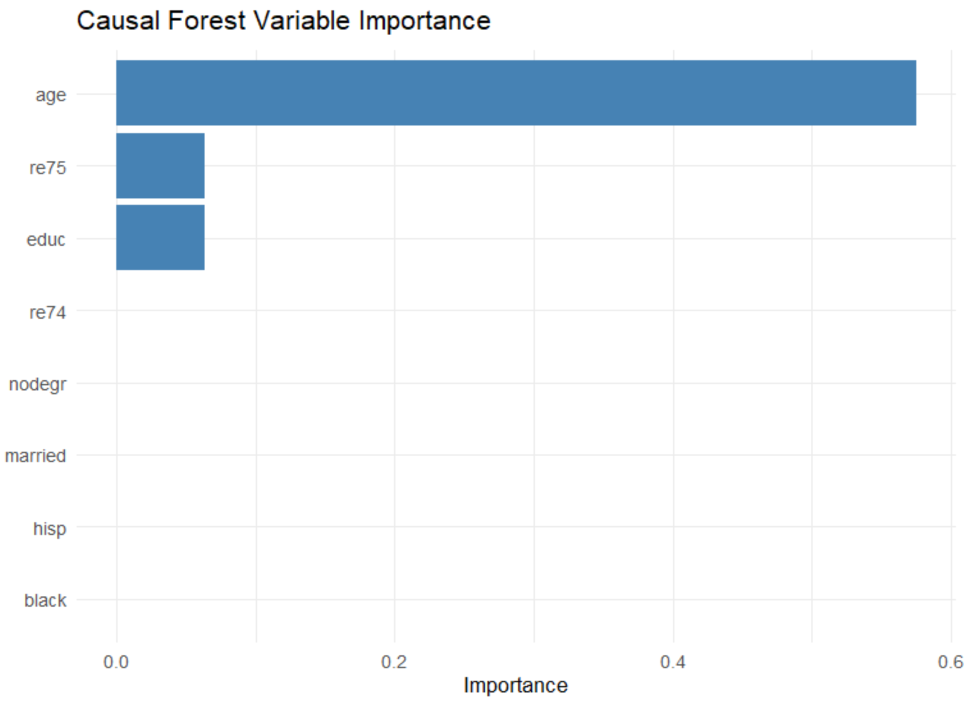

Conclusions

Consistent ATE ~$1,400–$1,800

All 8 methods agree the training program raised earnings. No method finds a zero or negative ATE.

PSM outlier at $2,626

ATT vs ATE distinction — treated group shows higher returns, consistent with selection into training.

Substantial heterogeneity

Causal Forest and BCF both detect significant CATE variation. Education and pre-earnings are top moderators.

Method convergence matters

Agreement across parametric and non-parametric methods strengthens causal credibility.